

Alternative Power

The 21st century brought pollution and fuel use into sharp focus. The industry responded with brilliantly innovative aircraft, from small fuel-efficient gasoline and diesel power plants, some running on environmentally friendly biofuels, to a wide range of electric aircraft (even carrying solar panels to maintain charge) and a hydrogen fuel cell-powered light aircraft. One pioneer promises Mach 1 solar flight in a decade.



△ Thruster T600N 450 2002

Origin UK

Engine 85 hp Jabiru 2200A air-cooled flat-4

Top speed 87 mph (140 km/h)

Introduced in the mid-1990s with a two-stroke engine, this is an enclosed-cockpit, side-by-side two-seat ultralight with an efficient, modern four-stroke engine.



△ Diamond DA40 TDI Star 2002

Origin Austria

Engine 135 hp Thielert Centurion 1.7 turbocharged liquid-cooled 4-cylinder inline diesel

Top speed 144 mph (232 km/h)

Austrian light aircraft-builder Diamond was the first to use this turbo diesel engine, developed from the Mercedes-Benz A-class 170 car unit. It proved unreliable, so Diamond sourced its own unit.



△ Lange Antares 20E 2003

Origin Germany

Engine 57 hp Lange EA42 electric motor

Top speed N/A

Lange Aviation offers self-launching gliders capable of taking off and climbing 10,000 ft (3,048 m) on battery power alone, with lithium-ion batteries in the wings and a built-in charging system.



△ Diamond DA42 2009

Origin Austria

Engine 2 x 168 hp Austro AE 300 turbocharged liquid-cooled 4-cylinder inline diesel

Top speed 222 mph (357 km/h)

Eco-aware Diamond began building its own diesel engines in 2008: the first flew in 2009. The DA42 was the first diesel plane to cross the Atlantic and in 2010 the first to fly on algae-derived biofuel.



△ Solar Impulse 2009

Origin Switzerland

Engine 4 x 10 hp electric motor

Top speed 43 mph (69 km/h)

Conceived by Bertrand Piccard, this solar-charged craft has four electric motors under its 747-scale wings. It has stayed aloft for 26 hours and is to be flown around the world.



△ Boeing-FCD 2008

Origin Spain

Engine Electric motor plus fuel cell

Top speed N/A

Boeing's Fuel Cell Demonstrator achieved the first fuel cell-powered level flight in February 2008. It was built with the aid of British, Austrian, US, and Spanish companies.